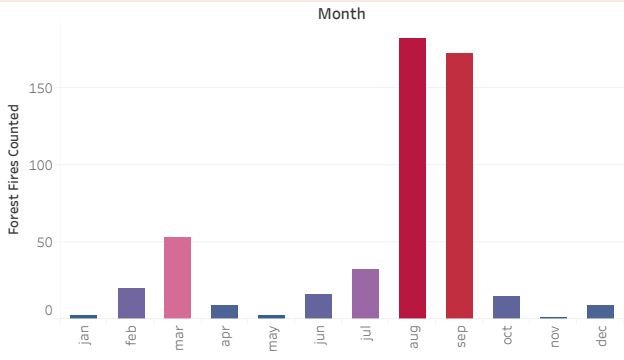


Forest Fire Analysis: Predicted Burned Area in Northeast Portugal

Phase 2 TEPP group project by Amir Benston, Lofinda B. Beynis and Palo Becerra, August 2026.

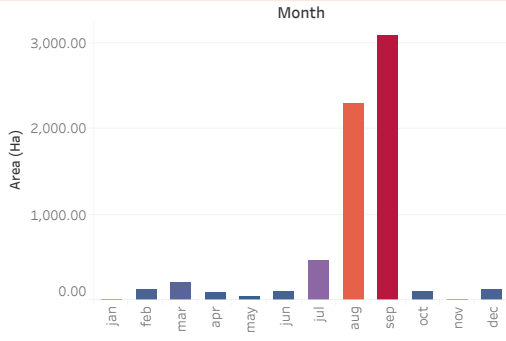
Forest Fires by Month: August + September Have Most Activity

Late summer weather tends to be hot and dry in the mountainous region of Northeast Portugal.



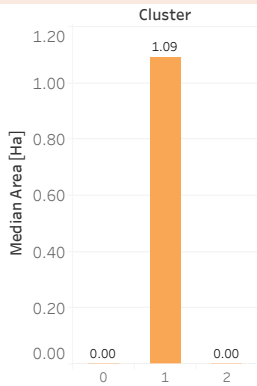
Burned Area by Month: September Peaks

Despite having fewer forest fires than August did, September consistently rose ahead as having the most burned area (3,086).



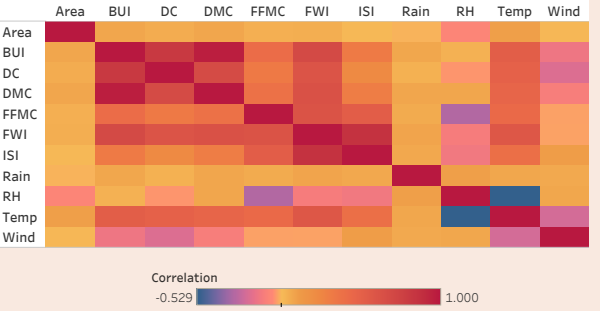
One Cluster Stood Alone: Fires' Habitual Spread

Cluster 1 was the only group where measurable fire spread was the typical outcome.



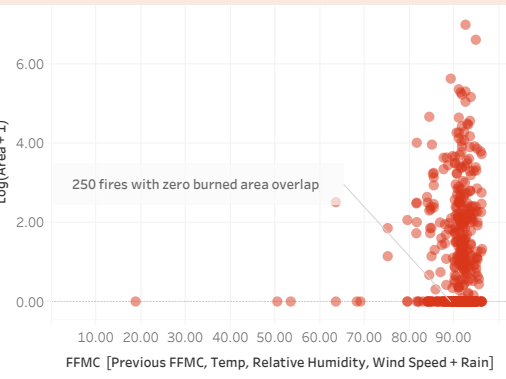
Correlation Heatmap: No Strong Predictors Re: Fire Size

No single weather variable or index correlated strongly with burned area. The strongest was temp at a mere 0.10.



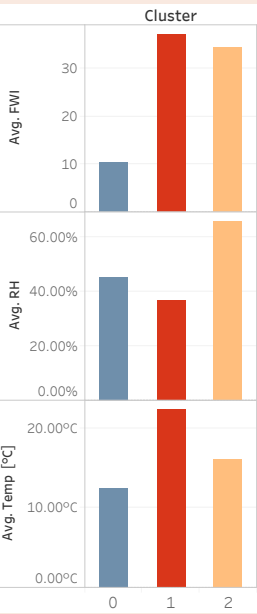
FFMC vs Log(Area + 1): Standout Predictor Was FFMC Index

Amongst all the raw variables, a composite feature was the only meaningful clue for further analysis.



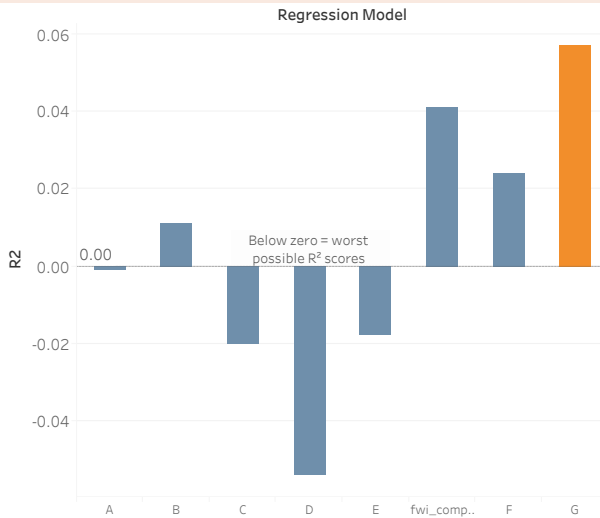
K-3 Clusters: Subtle Signals to Prediction

Patterns key to usage in Model G - the best performing model during analysis.



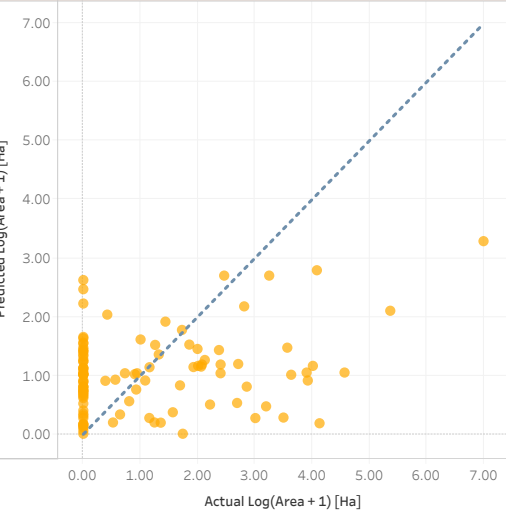
Eight Models, One Modest Win: Model G

No single variable, index, or model reliably predicted fire size. Grouped risk clusters came closest.



Predicted vs Actual: Model G

With such a difficult dataset, even the best model underpredicted the largest of forest fires.



Sources:

<https://archive.ics.uci.edu/dataset/162/forest+fires>

https://github.com/lofind-anyc/Forest_Fire_Analysis/tree/main